PNAS RESEARCH ARTICLE | Collective behavior

From hunter to shepherd: does the optimal alignment of a fish school depend on the agent that perturbs it?

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Introduction

A fish school under attack does not behave like a collection of independent prey. Its members turn, split and regroup together, and the predator faces a moving, changing target. A popular explanation for how such groups process information is the criticality hypothesis: living collectives would operate close to a phase transition, where their response to perturbations is largest (1, 2). For moving groups the relevant transition is the one between a disordered swarm and an ordered, polarized school, which appears in models of self-propelled particles as soon as alignment dominates noise (3, 4).

Klamser and Romanczuk (5) put this hypothesis to the test with a spatially explicit model in which a school of 400 prey is attacked by a single predator. The capture rate is indeed minimal near the order–disorder transition, but not for the reason the hypothesis suggests. When the attacked school is compared with an identical control school that does not flee, the share of prey saved by fleeing grows steadily with alignment and shows no peak at the transition. The advantage of the critical school therefore comes from its spatial structure, which is looser and more spread out, and not from a better collective response. Moreover, when alignment evolves under individual-level selection, the prey move far into the ordered phase, so the critical point is not evolutionarily stable.

These conclusions rest on one particular predator. It swims twice as fast as its prey, steers towards a weighted center of the prey in front of it and never changes its strategy. Real predators choose the direction of their attack (6), combine tactics such as dispersing a school before isolating a prey (7), and are confused when many prey move around them (8). It is not known whether the alignment that best protects a school depends on how it is attacked. The opposite question is also open. Responsiveness at criticality does not explain the low capture rate, but it may be exactly what an agent needs to steer a school without breaking it apart, as a sheepdog does with a flock (9) or a group of robots with a school of fish (10). Such guiding agents are being considered to lead fish away from dangers or along a given path (10, 11).

We therefore ask whether the benefit of operating near the order–disorder transition depends on the agent that perturbs the school. The project has three steps. First, we reimplement the model of Klamser and Romanczuk in Python and reproduce its main result. Second, we give the predator one behavioral improvement and test whether the result holds in three dimensions. Third, we turn the predator into a shepherd, whose goal is to move the school rather than to break it. This first report reviews the existing models, describes our reimplementation and its validation against the original code, and presents the reproduction of the main result.

Related work

Methods

Results

Discussion

Code availability

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